

DCG ReID: Disentangling Collaboration and Guidance Fusion Representations for Multi-modal Vehicle Re-Identification

Aihua Zheng, Ya Gao, Shihao Li, Chenglong Li*, *Senior Member, IEEE*, Jin Tang

Abstract—Multi-modal vehicle Re-Identification (ReID) aims to leverage complementary information from RGB, Near Infrared (NIR), and Thermal Infrared (TIR) modalities to retrieve the same vehicle. The challenges of multi-modal vehicle ReID arise from the uncertainty of modality quality distribution induced by the inherent discrepancies across multiple modalities, which results in distinct yet conflicting fusion requirements for data with balanced and unbalanced quality distributions. Existing methods handle all multi-modal data within a single fusion model, which causes the model to overlook the different needs of the two types of data, making it difficult to decouple the conflict between intra-class consistency and inter-modal heterogeneity. To this end, we propose to Disentangle Collaboration and Guidance Fusion Representations for Multi-modal Vehicle ReID (DCG-ReID). Specifically, to disentangle modal data with heterogeneous quality distributions while avoiding mutual interference, we first propose the Dynamic Confidence-based Disentangling Weighting (DCDW) mechanism, which realizes the disentangled fusion framework by dynamically reweighting the contributions of the three modalities via modal confidence derived from multi-modal interaction. Building on the DCDW, we further develop two scenario-specific fusion strategies to tackle distinct quality distribution characteristics: (1) for balanced multi-modal quality distributions, we introduce the Collaboration Fusion Module (CFM), which mines and extracts consensus features in a pairwise manner to capture shared discriminative information, in turn effectively enhances intra-class consistency; (2) for those with unbalanced quality distributions, we propose a Guidance Fusion Module (GFM), which implements differential amplification of modal discriminative disparities to reinforce the advantages of the dominant modality while guiding auxiliary modalities to mine complementary discriminative information, thereby effectively mitigating inter-modal divergence and substantially boosting multi-modal joint decision-making performance. The extensive experiments on three multi-modal vehicle ReID benchmarks WMVeID863, MSVR310 and RGBNT100 validate the effectiveness of our method. The code will be released upon acceptance.

Index Terms—Multi-Spectral Vehicle Re-Identification, Multi-modal Interaction, Multi-modal Confidential Fusion.

I. INTRODUCTION

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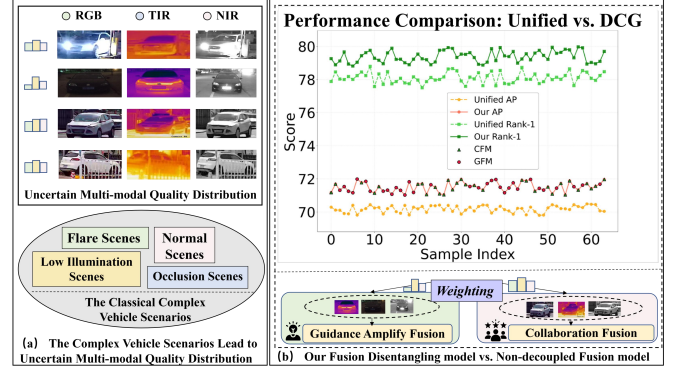


Fig. 1: (a) Inherent key challenges in multi-modal vehicle ReID lead to unbalanced multi-modal quality distribution. (b) The heatmap of AP and Rank-1 gains on samples between our fusion decoupling method and the unified fusion model (i.e., it does not distinguish between the quality distributions of multi-modal samples); The basic approach of our DCG based on fusion decoupling.

VEHICLE ReID task involves retrieving the same vehicle targets from non-overlapping camera views. Over the past decade, the recognition accuracy of ReID models has been continuously improving. Traditional vehicle ReID tasks are single-modal tasks, which use visible light image (RGB) for retrieval. Although RGB provides rich discriminative color cues [1, 2, 3, 4, 5, 6, 7, 8], single-modal vehicle ReID remains unreliable in challenging scenarios such as nighttime, low illumination, and glare. By contrast, TIR alleviates low-light limitations through thermal radiation perception, while NIR offers advantages under occlusion, flare contamination, and poor lighting. Therefore, introducing TIR [9] and NIR [10] when RGB lacks sufficient discriminative information has substantially advanced vehicle ReID. Zheng *et al.* [11] and Li *et al.* [10] are among the first to introduce multi-modal vehicle ReID and establish high-quality benchmarks. Since then, a growing body of research has leveraged multi-modal data to improve ReID performance, ranging from fine-grained alignment [12, 13, 14] to coarse-grained interactive fusion [15, 16, 17, 18, 19, 20]. Zheng *et al.* [9] utilize the introduced flare-immune TIR modality and fusing it with the other two modalities, which enhances the recognition performance under the challenge of nighttime flares.

Although multi-spectral technology enhances recognition accuracy in ReID tasks, it nevertheless brings about new challenges [10, 11, 21] regarding multi-modal alignment and fusion caused by modality discrepancies. As shown in Fig.

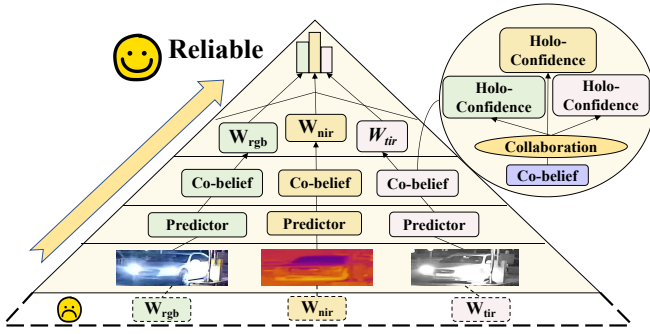


Fig. 2: Our reliable multi-modal dynamic weighting vs. single-modal weighting.

1 (a), these challenges stem from the unbalanced quality distributions across modalities caused by issues like low light, occlusion, and flare interference. To address these, existing methods use unified fusion models, such as multi-modal complementary alignment and dynamic fusion. For instance, Wang *et al.* [15] use the Mixture-of-Experts (MoE) fusion method to fuse single-modal, dual-modal, and tri-modal information. Zheng *et al.* [21] dynamically select the dominant modality to guide the learning of auxiliary modalities within a single fusion framework, significantly improving ReID performance on unbalanced quality datasets. Lin *et al.* [20] present DMPT, a decoupled prompt tuning framework for multi-modal ReID. Although it introduces decoupling operations, it still performs fusion within a unified single framework. As shown in Fig. 1 (b), which compares the ReID evaluation (AP and Rank-1 [22]) of each sample between the structurally coupled single fusion model and our fusion-decoupling model. The heatmap color denotes the Rank-1 gain, while the vertical difference reflects the AP gain per sample. It is evident that for multi-modal fusion with uncertain quality distributions, performing fusion decoupling first and then conducting targeted fusion based on quality characteristics is more effective. The core issue with structurally coupled fusion frameworks is that they overlook the inherent fusion conflict between modalities quality distributions. When the model excessively focuses on dominant modal, it will overfit to the unbalanced distributions reducing intra-class consistency. Conversely, when the model places more emphasis on balanced-distribution, it will neglect the advantageous features of the high-quality modality, leading to unexplored inter-modal heterogeneity.

Traditional dynamic fusion mainly relies on weight adjustment to handle uncertain modality quality distributions. When the quality of a certain modality drops sharply (e.g., RGB completely fails due to flare), the error-correcting capability of weight adjustment is limited, which easily leads to fusion results being dominated by noise. Accurately evaluating the dynamic characteristics of modality data quality has become the first key to breaking through fusion conflicts and further facilitating the full utilization of multi-modal information for final ReID decision-making. Inspired by trustworthy fusion applications derived from trustworthy learning [23], relevant studies [24, 25] on multi-modal trustworthy learning have shown that multi-modal learning can sometimes underperform

compared with partial-modal or even single-modal learning, especially when there is an imbalance in modality quality. In corresponding downstream tasks, Qu *et al.* [26] have enhanced the performance of multi-modal fusion by leveraging trustworthy weight reallocation. **To resolve modality fusion conflicts and enhance the positive gains of multi-modal fusion, we propose the Dynamic Confidence-based Disentangling Weighting (DCDW) fusion decoupling module** to differentiate multi-modal vehicle data with balanced and unbalanced quality distributions to avoid fusion conflicts. As shown in Fig. 3, DCDW calculates the mono-confidence and holo-confidence of each modality through multi-modal interaction and integrates them into a unified co-belief score to determine the final weights, thereby balancing modality contributions, identifying potential fusion conflicts, and facilitating targeted fusion.

After obtaining the modal weight distribution in this way, we correspondingly decouple the two fusion frameworks to make targeted decisions, respectively. As shown in the bottom of Fig. 1 (b), based on this confidence-based decoupled fusion mechanism, **we design two fusion strategies, namely the Guidance Fusion Module (GFM) and the Collaboration Fusion Module (CFM), for data with different quality distributions, thereby fully exploiting multi-modal data while avoiding fusion conflicts.** For balanced distribution, we leverage CFM to employ a dropout collaborative learning approach which is based on confidence-based weights via a bidirectional enhancement attention mechanism. It maximizes consensus information extraction to expand the intra-class consistency by conducting pairwise mining between modalities. Meanwhile, for unbalanced distributions, we design a GFM that leverages differential amplification with learnable factors. It amplifies unique information of the dominant modality while guiding auxiliary ones, thus reducing inter-modal heterogeneity through enhanced interactive learning. Thus, we leverage such multi-modal weights to discard low-quality modal samples, avoiding the interference caused by inferior modal data. In the case of unbalanced multi-modal quality distribution, we amplify the advantages of the dominant modality and guide the learning of auxiliary modalities, thereby enhancing the reliability of multi-modal joint decision-making.

In summary, the contributions of our work are as follows:

- To avoid mutual interference in fusion decision-making for uncertain quality multi-modal data distributions, we propose a Dynamic Confidence-Based Disentangling Weighting (DCDW) to decouple fusion based on adaptive confidence-based modal weights.
- We propose a Collaboration Fusion Module (CFM) to process modal features with balanced multi-modal quality distributions, mining shared information through pairwise inter-modal interactions and leveraging complementary shared information to expand intra-class consistency.
- We propose a Guidance Fusion Module (GFM) for scenarios with unbalanced multi-modal quality distributions. On the basis of amplifying the advantageous parts of the dominant modality, it guides the learning of auxiliary modalities, while reducing inter-modal differences and enhancing the effectiveness of multi-modal interactive

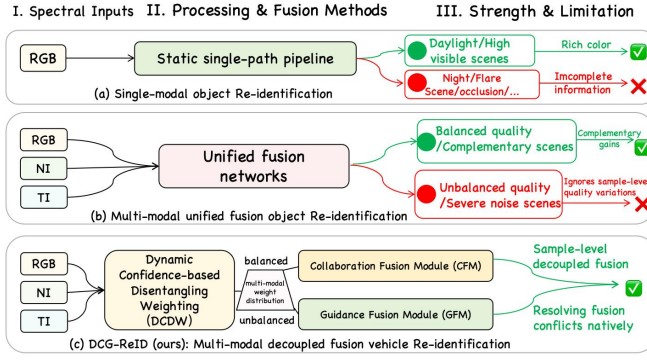


Fig. 3: Evolutionary comparison of vehicle ReID paradigms. (a) Single-modal object ReID [27, 28, 29]. (b) Multi-modal unified fusion networks [10, 11, 15, 17, 18, 19, 30] leverage multiple spectral inputs to achieve complementary gains; however, their rigid structure ignores sample-level quality variations, leading to suboptimal performance in unbalanced scenarios. (c) DCG-ReID introduces a multi-modal decoupled fusion architecture. By utilizing DCDW to assess weight distributions, the network dynamically routes samples into either the CFM or GFM, thereby resolving fusion conflicts natively at the sample level.

mining.

- Extensive experiments on three multi-modal vehicle ReID datasets WMVeID863 [9], MSVR310 [11] and RGBNT100 [10] validate the effectiveness of our method.

II. RELATED WORK

A. Multi-modal Vehicle ReID

Compared to single modality, introducing multi-modal improves the model's recognition robustness. As shown in Fig. 3, multi-modal enhancement is used in various recognition tasks, such as remote sensing [31, 32, 33, 34] and autonomous driving [35, 36]. In the closely related field of cross-modality pedestrian retrieval, Yu *et al.* [37] propose X-ReID to capture multi-granularity temporal information and handle visible-infrared gaps via prototype collaboration. In vehicle ReID, early studies mainly rely on RGB images, which are vulnerable to low light, occlusion, and flare. Unlike RGB images, NIR can capture clear vehicle details such as license plate edges and headlight shapes even in dim environments, and TIR is immune to complete darkness and zero-visibility weather. To address this limitation, multi-modal vehicle ReID introduces RGB, NIR, and TIR data. As NIR and TIR have unique imaging features offering new solutions, Li *et al.* propose RGBN300 [10] dataset and construct a new baseline method HAMNet to fuse spectra features to enhance ReID robustness. Zheng *et al.* [11] introduce a cross-directional network CCNet with a novel cross-directional center loss to simultaneously overcome the problems of cross-modality discrepancy and propose a benchmark dataset MSVR310.

However, the introduction of multiple modalities also brings the problem of imbalanced modality quality distributions. To reduce modality discrepancies and improve multi-modal gains, researchers have explored methods such as modality alignment

and modality fusion. Another line focuses on suppressing background noise and enhancing feature discrimination. Zhang *et al.* [12] propose EDITOR, which adopts a fine-coarse framework with a spatial-frequency token selection module to retain discriminative tokens and aggregate informative features. Zheng *et al.* [9] further construct the first flare-affected multi-spectral vehicle ReID dataset and exploit the robustness of the TIR modality under flare interference to enhance representation quality. Beyond discrimination enhancement, some methods emphasize modality interaction and information preservation. Wang *et al.* [15] combine dynamic weighting with multi-expert fusion to handle uncertain modality quality, while Yu *et al.* [13] design an Attention-Fourier Token Sparsification module together with an information unification constraint to preserve spectrum-specific cues while aligning multi-spectral features. Similarly, Liu *et al.* [30] introduce a selective interaction and global-local alignment network to extract discriminative features. In addition, some studies target specific scenarios or learning settings. Jiao *et al.* [38] propose a Pose-adaptive and Saliency Attention Network for the ReID of aerial-view vehicles to address deformation of features and invisible salient attributes caused by viewpoint changes. In the unsupervised setting, Qiu *et al.* [39] exploit intra-camera feature distributions to optimize pseudo-label generation and further improve pseudo-label utilization through focal contrastive learning. Furthermore, prompt-based paradigms [40] have emerged to bridge cross-modal gaps and compose semantic prompts for capturing unique object characteristics. While these methods have improved the recognition accuracy of multi-modal vehicle ReID models from different perspectives, they perform fusion within a unified single framework, risking conflict between intra-class consistency and inter-modal heterogeneity when modality quality distributions are different. In particular, performing full tri-modal fusion in all situations can be sub-optimal. In contrast, our proposed DCG utilizes fusion decoupling dynamically adjusts the fusion strategy according to confidence-based modal weights. Moreover, on the basis of avoiding fusion conflicts, our method designs two fusion approaches respectively targeting uniform and non-uniform multi-modal quality distributions.

B. Uncertainty in Deep Learning

Uncertainty estimations [41, 42, 43, 44] have been introduced to deep learning models to identify areas where networks are error-prone. Lou *et al.* [45] score uncertainties and propose a fusion module that exploits the Mixture-of-Expert architecture to encode multi-modal uncertainties in fusion pipelines to better percept autonomous driving. In the domain of object ReID, Zheng *et al.* [46] reweight the contribution of losses based on uncertainty measured by output consistency; Yang *et al.* [47] optimize the ReID model by selecting credible IDs to correct the noise in pseudo-labels. Furthermore, Zhao *et al.* [48] model image and text features as Gaussian distributions and estimating their multi-granularity uncertainty via batch-level and identity-level variances which acted as a feature augmentation and provided richer image-text semantic relationship. The introduction of multi-modal is

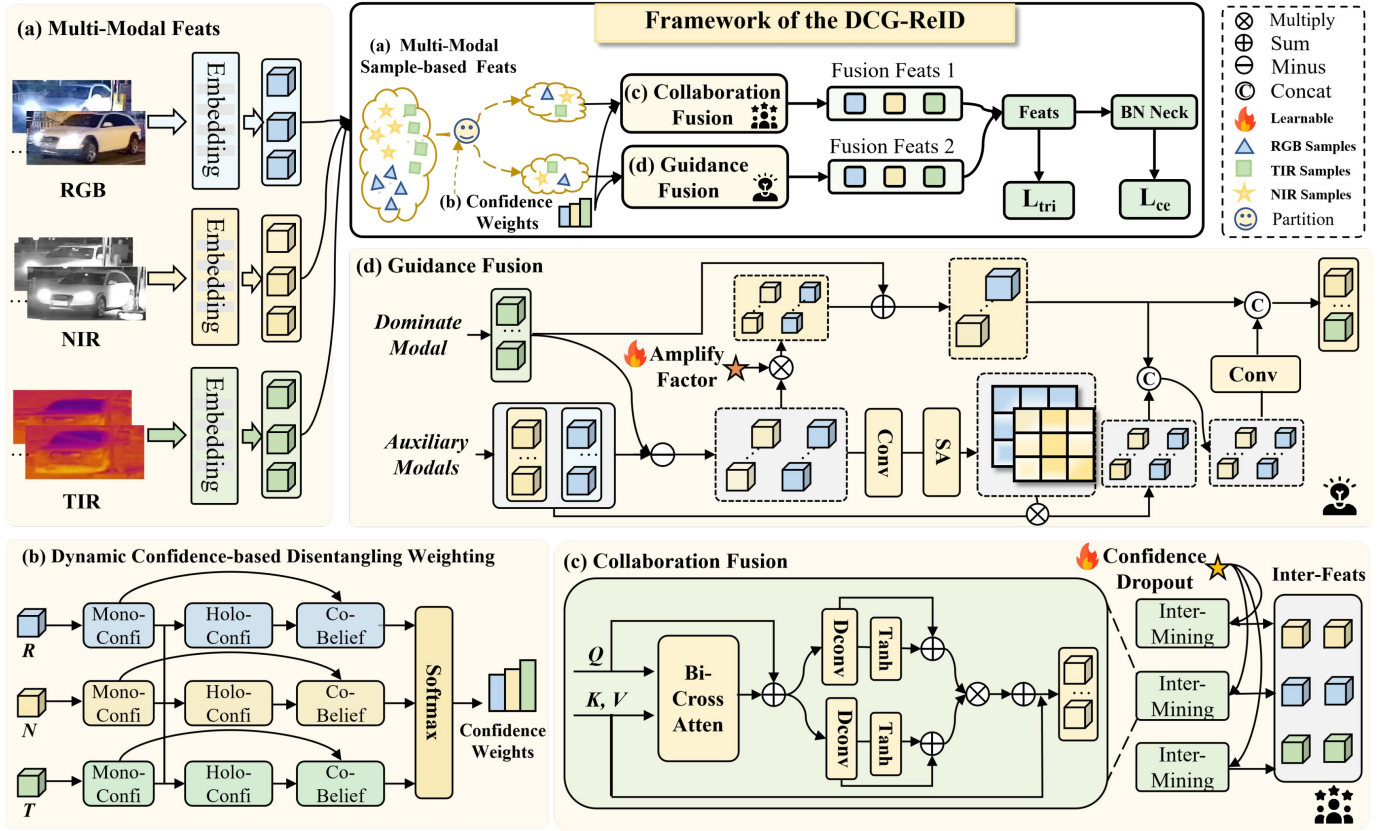


Fig. 4: The framework of our DCG. Firstly, the three-modal features (RGB, NIR, TIR) are extracted by the shared CLIP visual encoder. They then enter our designed Dynamic Confidence-based Disentangling Weighting (DCDW) to obtain dynamic confidence-based weights. According to the weight distribution, we introduce two fusion strategies, namely the Collaboration Fusion Module (CFM) and the Guidance Fusion Module (GFM), which are based on the reliable weights of the three modalities. Finally, we fuse these features to generate robust multi-modal features.

accompanied by the problem of multi-modal fusion. When researching multi-modal fusion methods, researchers should first focus on whether multi-modal fusion decisions are necessarily more effective than the previous single-modality or dual-modality approaches. Qiu *et al.* [24] indicate the importance of multi-modal learning in evaluating modality weights and thus measuring modality contributions when modality quality is unbalanced. Han *et al.* [49] introduce a uncertainty-aware fusion framework based on Dirichlet distribution and Dempster-Shafer Theory to enhance both classification reliability and robustness. These methods collectively demonstrate that modeling uncertainty not only improves robustness under modality corruption but also enhances interpretability in multi-modal fusion. Compared with previous methods for calculating confidence in ReID tasks, our method of learning credible fusion incorporates the credibility of multi-modal when computing the weight of each modality. Thus, the reliable weights provide a foundation for us to classify multi-modal data with uncertain quality distributions, conduct targeted fusion while avoiding fusion conflicts, and fully leverage multi-modal data to improve ReID performance.

III. METHODOLOGY

To enable the vehicle ReID model to reliably and adaptively select processing strategies for different complex real-world scenarios, we ensure the processing after adaptive selection is targeted to these scenarios. As shown in Fig. 4, we design DCG with three main components: Dynamic Confidence-based Disentangle Weighting (DCDW), Guidance Fusion Module (GFM) and Collaboration Fusion Module (CFM). We further introduce a sub-module, termed Inter-Mining, to enable cross-modal interaction and information mining. We will introduce these modules respectively in the following.

A. Feature Extraction

To effectively extract multi-modal features and reduce both modality differences and model parameters simultaneously, we employ a shared visual encoder from CLIP [50] pre-trained on ViT-B/16 to process the RGB, NIR, and TIR features.

$$F_m = \Theta(I_m), \quad (1)$$

where $I_m \in \mathbb{R}^{H \times W \times C}$ denotes the input image from modality m , and Θ represents the shared vision encoder. The output token features $F_m \in \mathbb{R}^{D \times (M+1)}$ are extracted from the final

layer of the encoder, where D is the feature dimension and M is the number of patch tokens.

The output feature F_m consists of a class token and a set of patch tokens:

$$F_m = [F_m^{cls}, F_m^{pat}], \quad (2)$$

where F_m^{cls} is the class token that captures global semantic information, and F_m^{pat} represents the local patch-level features.

B. Dynamic Confidence-based Disentangling Weighting

To differentiate fusion conflict data and assign a targeted fusion method to each sample more reliably, it is necessary to obtain reliable weights based on multi-modal interactions. In this case, we design a Dynamic Confidence-based Disentangling Weighting (DCDW) based on multi-modal confidence to comprehensively consider the reliable relationships among modalities.

We calculate two distinct confidence metrics for each modality, *Mono-Confidence* and *Holo-Confidence*, which accesses each modality's independent reliability and its collaborative reliability through multi-modal interactions, respectively. As Cao *et al.* [23] combining multiple confidence terms is more effective than using any individual term alone to represent modality weights. We comprehensively consider two confidence measures, denoted as *Co-Belief*, which reflects the confidence of every modality [51]. Therefore, we adopt this calculation approach when computing the weight of each modality.

Firstly, we utilize F_m^{cls} , which has global features, pay more attention to local features, we first use F_m^{cls} as Q and F_m^{pat} as K and V.

$$F_I = \Phi(Q_{cls}, K_{pat}), \quad (3)$$

where Φ denotes the Multi-head Cross-attention [52].

Then, we calculate the confidence of the inter modality itself as *Mono-Confidence* scores for three modalities are computed as:

$$M_{i,i \in R,N,T} = F_I \phi_i, \quad (4)$$

where $M_{i,i \in R,N,T}$ denotes *Mono-Confidence* _{$i,i \in R,N,T$} , ϕ_i are learnable parameters implemented with MLPs.

Secondly, we utilize *Holo-Confidence* to extend mono-confidence, rebalancing the contributions of the multi-modal. The *Holo-Confidence* for each modality is computed as:

$$H_{i,i \in R,N,T} = \frac{\sum_{j \neq i}^N M_{i,i \in R,N,T}}{\sum_{i=1}^N M_{i,i \in R,N,T}}, \quad (5)$$

where $H_{i,i \in R,N,T}$ denotes *Holo-Confidence* _{$i,i \in R,N,T$} , j identifies the current modality, and $N = 3$.

The coherent belief *Co-belief* incorporated two confidences is computed as:

$$C_i = M_i + H_i, \quad (6)$$

where C_i denotes *Co-belief* _{i} .

Finally, the dynamic multi-modal weights for each of the three modalities w_R , w_N and w_T are computed as:

$$w_{i,i \in R,N,T} = \text{softmax}(C_i). \quad (7)$$

Acting as an instance-level quality gatekeeper prior to feature aggregation, DCDW interacts multi-modal tokens to evaluate each modality's independent reliability (*Mono-Confidence*) and collaborative credibility (*Holo-Confidence*) into a unified *Co-Belief* score. After obtaining multi-modal weights, it is possible to dynamically differentiate data with uncertain quality distributions for subsequent targeted interactive fusion without fusion conflict, where the confidence mechanism is learned during training through modality interactions and applied during testing to assess modality reliability for unseen data, ensuring consistent fusion decisions.

C. Collaboration Fusion Module

To enhance intra-class consistency when modal quality is relatively balanced, and mutually explore inter-modal features to amplify the gain of multi-modal interaction, we introduce Collaboration Fusion Module (CFM). In the experiments of ReID with missing modalities II, we found that in some cases, the performance metrics were higher when certain modalities were missing than when all modalities were present. We realized that in situations with varying challenges and uneven quality, forcibly fusing poor-quality modalities is not as effective as discarding them. Therefore, we design a sample-level approach to discard a certain poor-quality modality for a specific sample.

In certain case, one modality may exhibit severely degraded quality such as RGB under low-light or night scenarios, making tri-modal fusion may introduce interference rather than complementary information. To prevent this, we introduce a sample-level confidence-based dropout mechanism to adaptively prune inferior feature components. For the retained modalities, a Tanh function is applied to enhance feature contrast against severe illumination variations, followed by element-wise multiplication to lock onto mutually verified target regions. Consequently, this module effectively maximizes complementary multi-modal gains while expanding intra-class consistency.

So while conducting collaborative fusion, we introduce the modality retention mechanism:

$$\tau = \sigma(\zeta(W_{avg})), \quad (8)$$

where ζ denotes MLP, $W_{avg} = (w_R + w_N + w_T)/3$, w_R , w_N and τ are obtained from the DCDW, σ is the Sigmoid function (constraining $\tau \in [0, 1]$) and MLP models the statistical characteristics of modal weights. Instead of using a rigid fusion rule, we introduce an adaptive threshold to filter unreliable modalities, which helps prevent low-quality modal features from degrading collaborative fusion. We initialize τ to 0.25 and design it as a learnable parameter. Under the setting where the sum of the three-modal weights is 1, the model will not encounter a situation where all three-modal samples are discarded simultaneously.

A modality m is discarded if its confidence weight falls below τ :

$$\mathcal{D}(m) = \begin{cases} 1, & \text{if } w_m < \tau \quad (\text{discard}), \\ 0, & \text{if } w_m \geq \tau \quad (\text{retain}), \end{cases} \quad (9)$$

$$M_{retain} = m|D(m) = 0, \quad (10)$$

where M_{retain} denotes retained modalities.

Inter-Mining. Based on discarding low-quality samples, we design a bidirectional mining module named Inter-Mining to explore the consensus information among modalities to interactively mine inter-class modality features.

Specifically, each pair of the retained modalities acts as a query to mine feature information from the other modality. For each modality, we use F_m as query Q_m to mine the features of K_n , where $m \neq n$ and $m, n \in \{R, N, T\}$:

$$F_{m2n} = \Phi(Q_m, K_n), \quad (11)$$

where Φ denotes the Multi-head Cross-attention [52].

To enhance feature representation [53], the features are split into two branches for separate processing, which is equivalent to enhancing features from different perspectives F_1^{m2n} and F_2^{m2n} :

$$F_1^{m2n}, F_2^{m2n} = \text{Split}(\text{DWConv}_3(F_{m2n}), \text{dim} = 1). \quad (12)$$

Then, the Tanh function constrains the value of features in the range of $[-1, 1]$, enhancing the contrast of the features and improving the robustness of the features against changes in illumination [54].

$$F_1'^{mn} = T(D(F_1^{mn})) + F_1^{m2n}, \quad (13)$$

$$F_2'^{mn} = T(D(F_2^{mn})) + F_2^{m2n}, \quad (14)$$

where T denotes Tanh function, and D denotes DWconv.

In order to encourage interaction between the two enhanced branches and highlight regions that are discriminative in both branches, we perform element-wise multiplication on features from two perspectives.

$$F_{m2n}' = F_1'^{mn} \odot F_2'^{mn} + F_n, \quad (15)$$

where $\odot$ means element-wise multiplication.

Finally, we obtain the interactive mined features F_{m2n}' . We fuse these interaction results to enhance the multi-modal feature representation:

$$F_{all} = \zeta(\text{Concat}(F_{m2n}', F_{n2m}')), \quad (16)$$

where ζ denotes MLPs. Based on confidence-based dropout and inter-mining of useful features, we enhance intra-class consistency, thereby fully realizing the complementary multi-modal advantages.

D. Guidance Fusion Module

Under unbalanced quality distributions (e.g., midnight low-light or over-exposed scenes), a flat fusion paradigm would inevitably submerge the clean modality's cues into auxiliary noise. GFM resolves this by implementing asymmetric, uncertainty-aware semantic guidance. To leverage the advantages of multi-modal, the modality with superior imaging quality is used to guide the learning of auxiliary modalities, thereby enhancing intra-class consistency. To this end, we introduce a Guidance Fusion Module (GFM). Based on Bayesian uncertainty theory [41], this module strengthens the guidance

of the dominant modality and promotes complementary information interaction across different modalities. Specifically, if the multi-modal weight distribution is unbalanced and the confidence weight w_i ($i \in \{rgb, nir, tir\}$) of one modality exceeds a learnable threshold β , this modality is identified as the dominant modality, where w_i denotes the contribution weight of the i -th modality and β (initialized as 0.35) controls whether the dominant-guided fusion mechanism is activated.

Firstly, the dominant modality is selected by integrating epistemic uncertainty with modal weights:

$$\mathcal{M}_{dom} = \arg \max_{m \in \{R, N, T\}} (\mathcal{U}_e(m) \cdot W(m)) \quad (17)$$

where $\mathcal{U}_e(m)$ denotes the epistemic uncertainty of modality m (estimated via Monte Carlo dropout [41]), and $W(m)$ represents modal weights, adhering to Bayesian decision principles. Secondly, we compute spatial-aware discrepancy vectors:

$$d_{xy}^{conv} = \text{SSM}(\zeta(d_{xy})), \quad (18)$$

where ζ denotes Conv, $d_{xy} = F_{dom} - F_j$ (with j as auxiliary modalities), and SSM denotes spatial attention to emphasize discriminative patterns in unique features.

We set the factors to be learnable rather than fixed, since the dominance and reliability of different modalities vary across samples and scenes. These discrepancies are then amplified via learnable factors and we use F_{dom} as reference:

$$F_{dom}^{En} = F_{dom} + \alpha_{ij} \cdot d_{ij} + \alpha_{iz} \cdot d_{iz}, \quad (19)$$

where $\alpha_{ij}, \alpha_{iz} \in \mathbb{R}^D$ denote learnable amplification factors, initialized as 1.0 and updated adaptively during training. By maximizing d_{xy} , the model is enabled to implicitly separate shared and unique information, which is consistent with the invariance principle in representation learning [55].

Thirdly, cross-modal guidance propagates dominant modality information to auxiliaries. Enhanced auxiliary features are obtained via:

$$\text{Attn}_{j \leftarrow dom} = \text{Softmax} \left(\frac{\text{Cos}(F_{dom}^{En}, F_j)}{\sqrt{d}} \right), \quad (20)$$

$$F_j^{En} = F_j + \text{Attn}_{j \leftarrow dom} \odot d_{xy}, \quad (21)$$

where $\text{Cos}(\cdot)$ denotes Cosine Similarity [56], $\odot$ denotes element-wise multiplication, ensuring uncertainty-aware guidance.

Finally, multi-modal fusion aggregates interactive features:

$$\text{inter}_{j2i} = \zeta(\text{Concat}(F_j^{En}, F_i^{En})), \quad (22)$$

$$F_{All} = \text{Concat}(\text{inter}_{j2i}, \text{inter}_{k2i}, F_i^{En}), \quad (23)$$

where ζ denotes Conv, $i = \arg \max W(m)$, $\{j, k\} = \{R, N, T\} \setminus \{i\}$.

Through modality interactive mining based on amplifying the dominance of high-confidence modalities, it achieves comprehensive utilization of modal information, narrowing modality heterogeneity while enhancing intra-class consistency.

Algorithm 1 DCG framework decoupling fusion flow

```

1: Input: Multi-spectral training data  $F_{rgb}, F_{nir}, F_{tir}$ .
2: Parameter: An image encoder  $\gamma$ , learnable dropout pa-
   parameter  $\tau$ , parameter  $\delta$  (initialized as 0.35) for modality
   quality distribution determination, and amplification pa-
   rameter  $\alpha_{ij}$ .
3: Output: The final multi-spectral loss  $L_{total}$ .
4: for  $n = 1$  to epochs do
5:    $W_{rgb}, W_{tir}, W_{nir} = \phi(F_{rgb}, F_{nir}, F_{tir})$ 
6:    $H_i, i \in \{rgb, nir, tir\} = \frac{\log w_{rgb}}{\log w_{rgb} \log w_{nir} \log w_{tir}}$ 
7:   if  $w_i > \beta$  then
8:      $\triangleright f_{rgb}, f_{nir}, f_{tir} = \text{GFM}(F_{rgb}, F_{nir}, F_{tir})$ 
9:   else if  $w_{avg} < \delta$  then
10:     $\triangleright f_{rgb}, f_{nir}, f_{tir} = \text{CFM}(F_{rgb}, F_{nir}, F_{tir})$ 
11:   end if
12:   Calculate ID loss and triplet loss according to Eqs. (24)
   and (25).
13: end for

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E. Objective Function

During the training phase, the backbone features and the fusion features of our DCG are jointly supervised by identity loss [57] and triplet loss [58].

$$L_{ReID}(X) = L_{id}(X) + L_{tri}(X), \quad (24)$$

where X represents input features for supervision. Classification loss and triplet loss are widely used in multi-modal ReID to obtain intra-class compact and inter-class separable feature distributions. To simultaneously preserve the modality-specific information extracted by the backbone network and the joint information obtained after multi-modal fusion, we impose loss supervision on both. Finally, the overall loss for our framework is defined as:

$$L_{total} = L_{ReID}^{DCG}(f) + L_{ReID}^{CLIP}([f_R, f_N, f_T]). \quad (25)$$

The detailed training pseudo-code is shown in Algorithm 1.

IV. EXPERIMENT

In this section, we conduct detailed experiments on the proposed framework. First, we introduce the datasets, evaluation protocols, and implementation details (Sec.IV.A-B). Second, we conduct comparative experiments with the latest methods on vehicle datasets and we conduct comparative experiments with missing modalities (Sec.IV.C). Then, we perform ablation experiments (Sec.IV.D) including effects of key components and discussions of our modules. Finally we further analyze the components of the framework through visual analysis of the proposed method (Sec.IV.E).

A. Datasets and Evaluation Protocols

We conduct experiments on three publicly available multi-spectral vehicle ReID datasets MSVR310 [11], RGBNT100 [10], and WMVeID863 [9] which include three vehicle scenarios with different challenges.

WMVeID863 [9] contains 4709 image triplets of 863 IDs of 8 camera views, the number of image triplets of each vehicle

varies from 1 to 39. The datasets are randomly selected 575 IDs with 3314 image triplets for training and 288 IDs with 1309 image triplets for testing. The gallery samples consist of all 288 IDs with 1309 image triplets. The query samples are randomly selected from the 288 gallery samples, consisting of 205 IDs with 928 image triplets.

MSVR310 [11] contains 6,261 high-quality vehicle images, divided into 310 different vehicles, with 2,087 samples, each consisting of 3 spectra modalities. The training set includes 155 vehicles and a total of 1,032 samples. The gallery set contains 1,055 samples of the remaining 155 vehicles, while the query set consists of 52 randomly selected vehicles and 591 samples from the gallery set. These samples are captured at long time spans, covering 8 viewpoints around the vehicle and various challenges such as illumination change, shadow, reflection, and color distortion.

RGBNT100 [10] selected 100 vehicles and added 17,250 additional thermal-infrared images to form a three-spectra dataset. This dataset includes 8,675 image triples from 50 vehicles for the training set and 8,575 triples from the other 50 vehicles for the test gallery set. From the test gallery, 1,715 samples are selected to form the query set.

Evaluation Protocols To ensure fair experimental evaluation, we follow previous research methods. We use mean Average Precision(mAP) which measures the average precision of all queries, represented by the area under the precision-recall curve, providing a comprehensive assessment of recall and precision performance and Cumulative Matching Characteristics (CMC) at Rank-K ($K = 1, 5, 10$) to assess performance, commonly used in ReID tasks, to evaluate our approach.

B. Implementation Details

Our model is implemented with the Pytorch toolbox. We conduct experiments on one NVIDIA RTX 3090 GPU. We resize the images of each spectrum to 128×256 in vehicle datasets to maintain the aspect ratio of vehicle and use random horizontal flipping, padding with 10 pixels, random cropping, and random erasing [64] as feature enhancement strategies. We present trainable parameters and FLOPs for complexity analysis. We select ViT-B/16 as our visual backbone. We optimize the model via Adam: initial learning rates 2.5×10^{-4} for WMVeID863 [9], 1×10^{-5} for RGBNT100 [10] and 3.5×10^{-3} for MSVR310 [11].

C. Comparison with State-of-the-art Methods

Comparison on WMVeID863 [9]. As shown in the Table. I, we compare our method with the current state-of-the-art single-spectral and multi-spectral approaches. Notably, on the WMVeID863 [9] dataset, which introduces severe illumination interference caused by strong lens flares, our method DCG achieves the best performance across all evaluation metrics. To demonstrate the superiority of our method through comparison, we compare it with excellent methods specifically designed for vehicle challenges. Compared with FACENet [9], which enhances other spectras by leveraging the flare immune characteristics of the TIR spectrum, DCG outperforms it by **+1.6%** in mAP and **+3.1%** in Rank-1. However, in

TABLE I: Performance comparison on multi-modal vehicle ReID datasets. The best results are in **bold** and the second in underlined. The symbol $\dagger$ denotes CLIP-based methods, $^\circ$ denotes DINOv3-based, * indicates ViT-based methods, and others are CNN-based methods.

	Methods	Venue	RGBNT100		WMVeID863				MSVR310			
			mAP	R-1	mAP	R-1	R-5	R-10	mAP	R-1	R-5	R-10
Single	BoT [27]	CVPRW'19	78.0	95.1	51.1	55.7	69.8	74.7	23.5	38.4	56.8	64.8
	OSNet [29]	ICCV'19	75.0	95.6	42.9	46.8	61.9	69.4	28.7	44.8	66.2	73.1
	AGW [59]	TPAMI'21	73.1	92.7	30.3	35.3	43.3	46.5	28.9	46.9	64.3	70.7
	TransReID* [28]	ICCV'21	75.6	92.9	67.0	74.7	79.5	82.4	26.9	43.5	62.4	70.7
	DINOv3 $^\circ$ [60]	-	87.0	98.5	70.1	80.8	84.3	87.7	68.2	83.3	93.5	94.4
Multi-modal	HAMNET [10]	CVPR'18	74.5	93.3	45.6	48.5	63.1	68.8	27.1	42.3	61.6	69.5
	PFNet [61]	AAAI'21	68.1	94.1	50.1	55.9	68.7	75.1	23.5	37.4	57.0	67.3
	IEEE [62]	AAAI'22	61.3	87.8	45.9	48.6	64.3	67.9	21.0	41.0	57.7	65.0
	CCNet [11]	INFFU'23	77.2	96.3	50.3	52.7	69.6	75.1	36.4	55.2	72.4	79.7
	TOP-ReID* [14]	AAAI'24	81.2	96.4	67.7	75.3	80.8	83.5	35.9	44.6	-	-
	EDITOR* [12]	CVPR'24	82.1	96.4	65.6	73.8	80.0	82.3	39.0	49.3	-	-
	HTT* [63]	AAAI'24	75.7	92.6	66.2	73.2	79.9	82.3	34.5	43.2	-	-
	FACENet* [9]	INFFU'25	81.5	96.9	69.8	77.0	81.0	84.2	36.2	54.1	-	-
	DeMo $\dagger$ [15]	AAAI'25	86.2	97.6	68.8	77.2	81.5	83.8	49.2	59.8	-	-
	DeMo $^\circ$ [15]	AAAI'25	<u>88.4</u>	96.1	<u>72.5</u>	<u>79.9</u>	<u>85.2</u>	<u>89.2</u>	<u>68.7</u>	<u>81.6</u>	<u>90.7</u>	<u>93.1</u>
	Mambapro $\dagger$ [18]	AAAI'25	83.9	94.7	69.5	76.9	80.6	83.8	47.0	64.0	-	-
	IDEA $\dagger$ [19]	CVPR'25	87.2	96.5	-	-	-	-	47.0	62.4	-	-
	IDEA $^\circ$ [19]	CVPR'25	88.2	96.5	-	-	-	-	67.0	82.4	89.0	90.1
	PromptMA $\dagger$ [16]	TIP'25	85.3	97.4	-	-	-	-	55.2	64.5	-	-
	PromptMA $^\circ$ [16]	TIP'25	86.8	97.4	-	-	-	-	58.2	68.5	-	-
	CoEN $\dagger$ [21]	arxiv'25	86.4	96.6	70.9	77.9	82.7	85.6	52.2	69.5	-	-
	ICPL $\dagger$ [17]	TMM'25	87.0	<u>98.6</u>	67.2	74.0	81.2	85.6	56.9	77.7	87.6	91.5
	DCG $\dagger$	Ours	87.6	97.6	71.4	80.1	85.4	88.6	62.9	77.5	91.7	94.2
	DCG $^\circ$	Ours	88.0	98.9	72.4	82.8	86.4	89.3	69.3	85.2	93.7	94.8

cases where the quality of multi-modal is uneven, it is prone to reducing the discriminative power of the fused features when the RGB is contaminated by glare. Although CoEN considers using the dominant mode to assist in optimizing the secondary mode to enhance the model's recognition ability under the condition of uneven quality distribution, it has not yet addressed the situation where the three modes have balanced mass. These demonstrate the applicability of our DCG to uncertain modality quality distributions caused by complex vehicle scenarios.

Comparison on RGBNT100 [10] and MSVR310 [11]. On the large-scale RGBNT100 [10] dataset, IDEA [19] constructs a text-enhanced multi-modal object ReID benchmark, leveraging modal prefixes and inverse networks to integrate multi-modal features, further refreshing the performance to 87.2% mAP and 96.5% Rank-1. However, DCG continues to set new benchmarks by achieving 87.6% mAP and 97.6% Rank-1 on the RGBNT100 [10] dataset by decoupling fusion and two designed fusion strategies. When faced with the challenges of viewpoint changes and long time spans in the MSVR310 [11] dataset, common methods often experience performance degradation. In contrast, our DCG framework, designed specifically for complex and dynamic scenarios, utperforms DeMo and PromptMA on MSVR310 [11] by **+13.7%/+7.7%** in mAP and **+17.7%/+13%** in Rank-1. This fully demonstrates the exceptional effectiveness of our fusion framework in handling significant variations in viewpoint and temporal span.

For a fairer comparison, we reproduce a stronger baseline model DINOv3 [60], implement our method on top of it, and

also reproduced several state-of-the-art methods. As shown in the Table. I, compared with the reproduced advanced methods, including DeMo [15], IDEA [19], and PromptMA [16], our method DCG still achieves further improvement over the DINOv3 baseline, demonstrating its good adaptability to the baseline model. In summary, these results validate the superiority of the DCG method in the face of complex vehicle scenarios and large-scale vehicle datasets, proving that our method can address the main challenges of multi-modal vehicle ReID.

Multi-modal Vehicle ReID with Missing Modalities.

To assess DCG's robustness in missing-modality scenarios, as shown in Table. II, we conduct experiments on WMVeID863 [9], MSVR310 [11] and RGBNT100 [10]. By comparing with advanced methods, our performance remains state-of-the-art even when there are missing modalities. Despite lacking specific designs like the reconstruction modules in TOP-ReID [14] and PromptMA [16], our DCG achieves competitive performance through dynamic multi-modal fusion. In terms of average performance across the three vehicle datasets with different challenges, our method achieves the best results in both mAP and Rank-1. Specifically, on MSVR310, its mAP/Rank-1 are **+14.0%/+11.3%** higher than those of DeMo [15] respectively. On the WMVeID863 dataset with vehicle flare pollution, the average mAP/Rank-1 are improved **+6.4%/+8.2%** compared with DeMo. On the large-scale dataset RGBNT100 [10], the average performance of our method in terms of mAP/Rank-1 is **+2.8%/+0.5%** higher than that of DeMo. Combined with the experimental data of multiple missing scenarios, it shows that our method can

TABLE II: Performance of missing-modality settings on MSVR310 [11], WMVeID863 [9] and RGBNT100 [10]. “M(X)” means missing the X image modality. The best results are in **bold** and the second in underlined.

Methods	M (RGB)		M (NIR)		M (TIR)		M (RGB+NIR)		M (RGB+TIR)		M (NIR+TIR)		Average	
	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1
RGBNT100														
CCNet [11]	66.8	90.2	73.2	92.2	60.0	82.9	44.4	75.0	42.4	63.8	49.5	69.8	56.0	79.0
TOP-ReID [14]	70.6	90.6	77.9	94.5	64.0	81.5	42.5	69.3	45.9	65.4	55.4	77.8	59.4	79.9
DeMo [15]	81.0	94.1	<u>84.1</u>	96.5	71.1	<u>87.6</u>	<u>50.2</u>	73.7	59.6	78.1	<u>66.3</u>	<u>82.8</u>	<u>68.7</u>	<u>85.5</u>
PromptMA [16]	80.3	95.8	83.5	96.8	69.3	85.8	48.0	<u>74.3</u>	56.2	<u>75.2</u>	64.2	82.0	66.9	85.0
DCG (Ours)	81.7	<u>95.3</u>	87.1	<u>96.5</u>	<u>70.8</u>	87.8	51.1	76.2	<u>58.4</u>	74.8	69.1	85.4	69.7	86.0
MSVR310														
CCNet [11]	25.5	43.7	27.2	42.3	26.4	41.6	10.8	30.1	20.6	34.2	24.1	37.2	22.4	38.2
TOP-ReID [14]	23.5	41.1	28.6	41.8	31.1	45.9	10.8	23.5	22.3	40.6	26.5	36.5	23.8	38.2
DeMo [15]	<u>36.9</u>	<u>55.3</u>	<u>43.1</u>	<u>56.5</u>	<u>46.1</u>	<u>60.9</u>	<u>10.5</u>	<u>24.2</u>	<u>34.1</u>	<u>53.5</u>	<u>40.8</u>	<u>53.6</u>	<u>35.5</u>	<u>50.7</u>
DCG (Ours)	54.2	71.0	59.3	69.9	57.2	67.8	28.1	39.9	45.0	60.1	53.4	63.4	49.5	62.0
WMVeID863														
CCNet [11]	58.7	64.8	59.8	64.9	53.5	58.7	51.9	56.8	43.0	44.1	46.7	48.6	51.9	56.8
TOP-ReID [14]	63.2	69.9	<u>66.8</u>	<u>73.7</u>	55.3	<u>59.8</u>	<u>60.3</u>	<u>65.7</u>	46.8	<u>49.5</u>	<u>53.0</u>	<u>56.4</u>	57.6	62.5
DeMo [15]	<u>66.5</u>	<u>73.3</u>	65.9	<u>73.3</u>	<u>55.5</u>	<u>59.8</u>	58.4	63.9	<u>49.7</u>	52.1	50.4	52.8	<u>57.7</u>	<u>62.5</u>
DCG (Ours)	70.6	78.1	73.1	79.9	61.4	68.7	64.3	71.0	57.0	63.0	58.7	63.5	64.1	70.7

handle various modality missing problems well.

These results validate the superiority of the DCG approach in complex scenarios multi-spectral vehicle datasets, demonstrating its effectiveness over existing solutions.

D. Ablation Study

To thoroughly validate the effectiveness of each component of our proposed modules, we conduct ablation studies on the WMVeID863 [9] dataset on the Clip-based [50] backbone to validate the proposed components.

Effects of Key Components. Table. III shows the performance comparison with different modules. Model A is the baseline model. Model B and Model C introduce two fusion methods GFM and CFM respectively, increasing the mAP/Rank-1 **+3.9%/+4.8%** and **+3.7%/+4.6%**, which indicates the effectiveness of our two fusion strategies. The model B perfectly adapts to the characteristic of multi-modal data where a certain modality gains an advantage due to specific scene conditions, thereby verifying the effectiveness of the module in fusion. When the quality of multi-modal vehicle data is balanced, the information carried by each modality exhibits strong complementarity. The model C introduces CFM leverages confidence-based dropout to retain the core information of each modality while discarding redundant contour features. Subsequently, it employs bidirectional attention to enforce information alignment across modalities. We replace our DCDW selected weights by randomly feeding modal samples into the two fusion strategies formed Model D. Compared to our confidence-based differentiation method DCDW, the mAP/Rank-1 perform **-1.6%/-1.8%** and **-1.1%/-2.9%**, respectively. Model E combines the DCDW and two fusion strategies, achieving the best results with mAP **71.1%** and Rank-1 **80.1%**. These results clearly demonstrate the effectiveness and efficiency of our proposed modules.

Discussion on DCDW. As shown in Table. IV, we evaluate the effectiveness of DCDW. Specifically, when we randomly

TABLE III: Performance comparison with different modules.

Component				WMVeID863			
Index	DCDW	Fusion		mAP	R-1	R-5	R-10
		GFM	CFM				
A	×	×	×	66.7	74.2	80.6	82.3
B	✓	✓	×	70.6	79.0	84.7	86.7
C	✓	×	✓	70.4	78.8	83.5	85.9
D	×	✓	✓	69.8	78.3	84.0	86.5
E	✓	✓	✓	71.1	80.1	84.9	86.7

TABLE IV: Comparative discussion on dynamic sample partitioning methods of WMVeID863 [9] dataset. Among them, w/o (free) indicates that samples are randomly partitioned; w/o (TMC) indicates that we partition samples using the evidence theory [49] method; w/o (all) indicates that we treat samples uniformly and feed them into both fusion methods simultaneously.

Index	Method	WMVeID863			
		mAP	R-1	R-5	R-10
A	w/o (free)	69.8	78.3	84.0	86.5
B	w/o (TMC)	69.2	75.3	84.2	86.9
C	w/o (all)	70.3	77.2	82.0	85.1
D	w (ours)	71.4	80.1	85.4	88.6

select fusion strategies for samples (A), the mAP/Rank-1 scores decrease by **-1.6%/-1.8%** respectively. To explore other methods for evaluating sample quality in the application of ReID, as in Table. IV model B, we compare the TMC [49] which is combined with the proposed CFM or GFM module with our DCG. Our DCDW outperforms the evidence theory-based approach on WMVeID863 by leveraging both the modality itself and the interaction between modalities. TMC treats each modality as an independent evidence source to improve the accuracy of category decision, but DCG evaluates

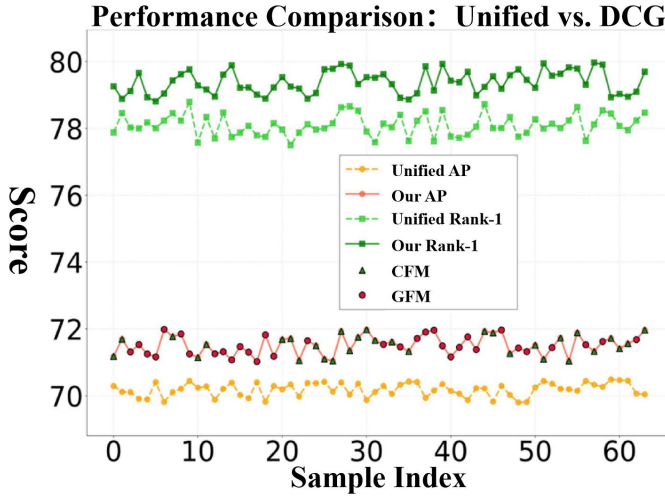


Fig. 5: Comparison of WMVeID863 performance on samples between our decoupling fusion method and unified fusion method (feeding all samples into two fusion strategies).

modal quality by integrating both the modalities themselves and their interactions to make fusion decisions. Our goal of using confidence assessment to evaluate the quality of modalities is to distinguish the quality distributions of multi-modal data. We feed all samples into two fusion strategies (C) to verify the necessity of fusion decoupling. When samples are subjected to two fusion measures simultaneously, compared with our fusion decoupling framework, the mAP/Rank-1 scores decrease by **-1.1%/-2.9%**, respectively. By comparing several different methods of assigning multi-modal data to fusion measures with our confidence-based reliable assignment, we can verify the reliability of our approach. The confidence-based dynamic allocation mechanism we designed for multi-modal data, which exhibits higher reliability in handling the quality distribution characteristics of multi-modal data.

TABLE V: Comparison of removing dropout (Drop) and differential amplification (Amplify) in CFM and GFM with the full model.

Methods	WMVeID863			
	mAP	R-1	R-5	R-10
w/o CFM (Drop)	70.8	78.3	82.0	85.1
w/o GFM (Amplify)	70.3	77.0	79.8	83.2
DCG (ours)	71.4	80.1	85.4	88.6

Discussion on Fusion Decoupling. As shown in Fig. 5, we use line charts and graphic markers to more clearly demonstrate the effectiveness of fusion and decoupling, while comparing our method with the one that does not adopt fusion and decoupling. For each multi-modal sample, the AP and Rank-1 values obtained when processing through both fusion methods simultaneously are lower than those achieved by our fusion decoupling approach. This directly demonstrates that our DCDW can avoid fusion conflicts and the necessity of fusion decoupling. By comparing several different methods of assigning multi-modal data to fusion measures with our confidence-based reliable assignment, we

can verify the reliability of our approach. The confidence-based dynamic allocation mechanism we designed for multi-modal data, which directs data into targeted fusion methods, exhibits higher reliability in handling the quality distribution characteristics of multi-modal data.

Discussion of CFM and GFM. As shown in the Table. V for CFM, we remove its dropout design, meaning that instead of discarding modalities with low weights, we treat all modalities equally and perform pairwise feature mining. Compared with DCG, mAP/Rank-1 perform **-0.6%/-1.8%** respectively, which verifies the rationality of our design in CFM to discard the modality with low weights. Based on a learnable discard factor, we can adaptively discard the low-quality modality, thereby avoiding the negative impact of this modality on the fusion of multi-modal data. As shown in Table. V for GFM, we remove the factor α in the GFM that amplifies the dominant modality, leaving the dominant modality with only a guiding function. When the quality of multi-modal data is unbalanced, there will be a dominant modality. The adaptive amplification mechanism of GFM can effectively amplify the advantages of the dominant modality, thereby enhancing the fusion performance of multi-modal data. As a result, mAP/Rank-1 perform **-1.1%/-3.1%** respectively, which proves that our differentiate amplification and guiding attention to the dominant modality through confidence-based weights is crucial.

Sensitivity Analysis of Hyper-parameters. Fig. 7 illustrates the effect of the key parameter in our CFM module on the experimental results over the MSVR310 dataset. This parameter is used to determine whether a modality should be regarded as the dominant modality to guide the learning of the other two modalities. In our DCDW module, the weights of the three modalities are constrained to sum to 1. We observe that, on this dataset, the weight of a single modality does not reach or exceed 0.4. Therefore, when this parameter is set to 0.4 or higher, the model performance remains unchanged. From these results, we can conclude that the optimal value of this parameter is 0.25.

Fig. 8 shows the sensitivity analysis of the key threshold in the GFM module on MSVR310. This threshold determines whether a modality is selected as the dominant modality to guide the others. As shown in the figure, the performance first improves and then declines as the threshold increases. The best result is achieved at 0.35, with 62.9% mAP and 77.5% Rank-1.

Discussion on the Bimodal Vehicle ReID Dataset. To analyze the effectiveness and generalization ability, we conducted a comparative analysis with existing SOTA methods on the RGBN300 [10] dataset. Compared with DeMo [15], our method achieves **0.9%/0.5%** improvements in mAP/R-1. This is because we do not perform simple multi-modal dynamic fusion, demonstrating that our fusion-then-decoupling strategy is also effective on bimodal datasets. Compared with PromptMA [16], we obtain **1.7%/0.9%** gains in mAP/R-1, as our approach pays more attention to the modality quality distribution. These experimental results verify the effectiveness and robustness of our method on datasets with different scales and numbers of modalities.

The Experimental Cost. Table VII presents a comparison

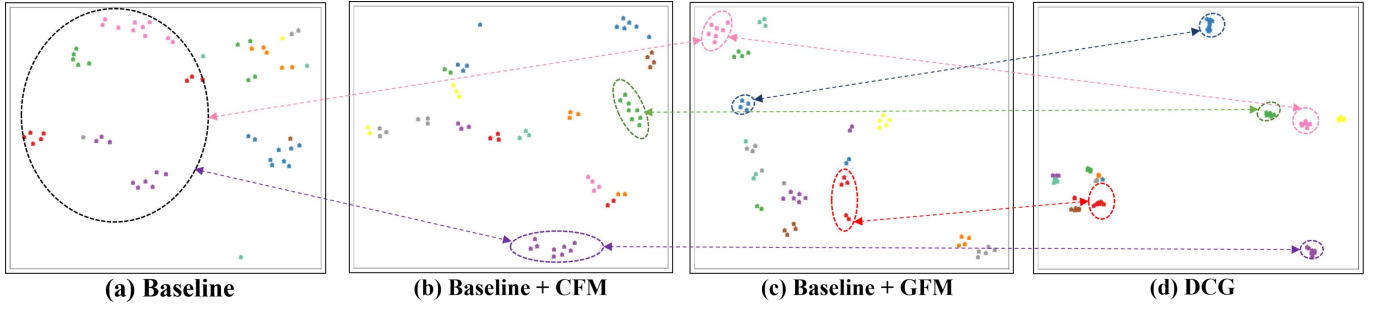


Fig. 6: Feature distributions with t-SNE [65] on WMVeID863. Different colors represent different identities. (a) Baseline, (b) Baseline + CFM, (c) Baseline + GFM, (d) DCG. Better view with colors and zooming in.

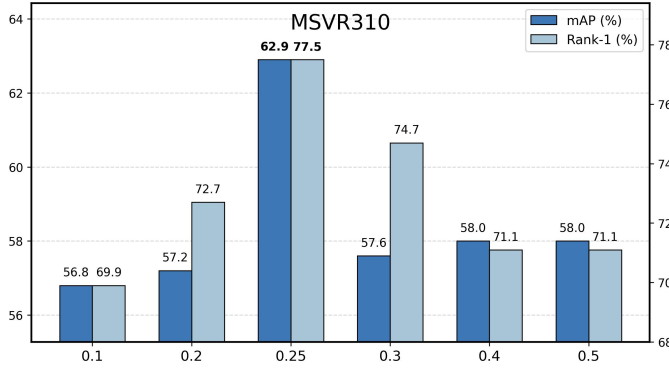


Fig. 7: Hyper-parameter analysis on factor τ .

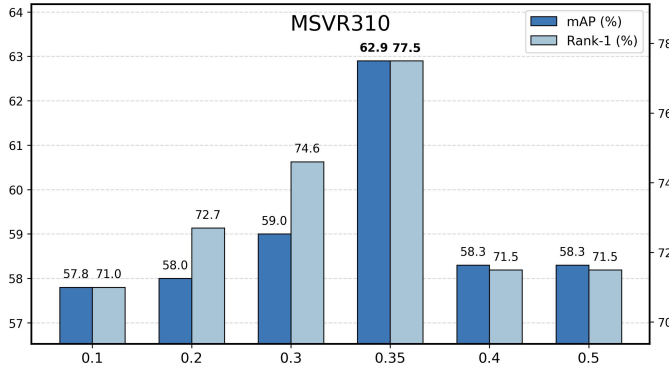


Fig. 8: Hyper-parameter analysis on factor β .

between our method and IDEA, both of which introduce an additional text branch during inference. It can be observed that, under the same backbone setting, our method achieves substantially better recognition performance with lower computational cost in terms of GFLOPs. Although DCG contains more parameters than IDEA, its lower FLOPs and significantly stronger performance indicate a more efficient use of model capacity. This verifies that our method is able to achieve a better balance between effectiveness and efficiency, highlighting its practicality for real-world multi-modal vehicle ReID applications.

E. Visualization Analysis

Multi-modal sample weight distribution. We employ the T-SNE [65] dimensionality reduction technique to visualize and

TABLE VI: Performance Comparison on RGBN300. The symbol $\dagger$ denotes CLIP-based methods and others are CNN-based methods.

Methods	Venue	RGBN300			
		mAP	R-1	R-5	R-10
HAMNET	AAAI'24	61.9	84.0	86.0	87.0
DeMo $\dagger$	AAAI'25	87.1	97.5	98.3	99.1
PromptMA $\dagger$	TIP'25	86.3	97.1	98.0	99.0
DCG $\dagger$	Ours	88.0	98.0	99.0	99.3

TABLE VII: Comparison of method performance in terms of model size, FLOP, mAP, and R-1. Symbols: * (ViT-based), $\dagger$ (CLIP-based).

Method	Paras (M) $\downarrow$	FLOP (G) $\downarrow$	mAP $\uparrow$	R-1 $\uparrow$
TOP-ReID*	324.5	69.1	35.9	44.6
EDITOR $\dagger$	119.3	40.8	39.0	49.3
PromptMA $\dagger$	107.4	67.4	55.2	64.5
DeMo $\dagger$	98.8	35.1	49.2	59.8
IDEA $\dagger$	91.7	43.7	47.0	62.4
DCG $\dagger$	131.3	34.4	62.9	77.5

analyze the feature distributions of vehicles. In the Fig. 6, we visualize the feature distributions of module ablations compared with the baseline. Compared with Fig. 6 (a) and (d), DCG further reduces the distance between instances of the same ID, making the features of each ID more compact and increasing the gap between different IDs. Compared with Fig. 6 (b)(c) and (d), features for each ID become more compact and the gap between different IDs increases from our two fusion strategies. It fully demonstrates that our proposed method can expand intra-class consistency while reducing inter-modal differences, thereby achieving excellent discriminative ability.

Class Activation Map Visualization. To validate the effectiveness of DCG, we visualize the channel activation maps in Fig. 9 and Fig. 10 on WMVeID863 [9]. To better demonstrate robustness in complex scenarios, we select two representative challenges, low light and flare pollution. In Fig. 9, “w/o All” means that all samples are simultaneously fed into both fusion strategies, while “w/o CFM” and “w/o GFM” denote the removal of the corresponding modules. The results show that CFM and GFM suppress background responses and

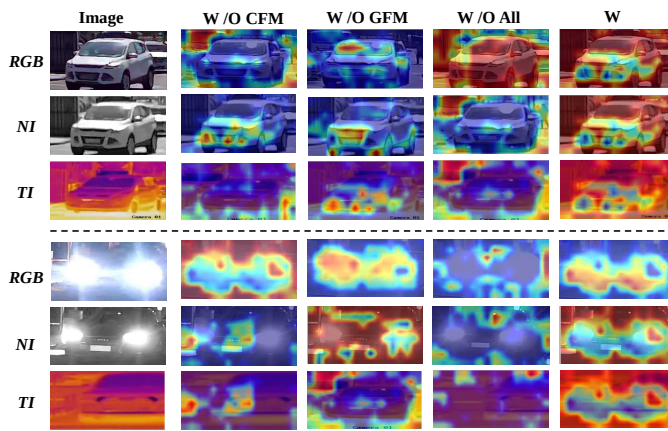


Fig. 9: Class activation map visualization on WMVeID863 from different methods with two main challenges.

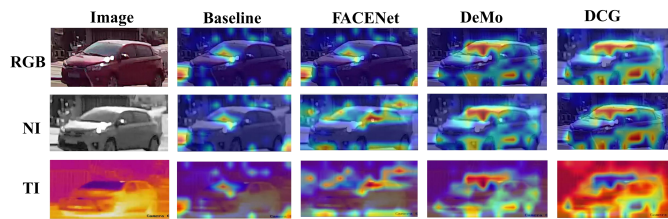


Fig. 10: Class activation map visualization on WMVeID863 comparing different methods.

enhance vehicle-focused activations, especially in the infrared modality. These comparisons verify that our fusion-decoupling framework effectively avoids fusion conflicts and enables the model to focus more accurately on the vehicle itself under challenging conditions.

V. CONCLUSION

In this paper, we present a novel framework named DCG for multi-modal vehicle ReID. Our approach starts with a Dynamic Confidence-based Disentangling Weighting (DCDW) to differentiate multi-modal data with different quality distributions without fusion conflicts. Then, we introduce two fusion strategies named Collaboration Fusion Module (CFM) and Guidance Fusion Module (GFM) to achieve targeted fusion for uncertain modality quality distribution data. Among the two fusion methods, we respectively leverage adaptively and discard modalities with poor quality confidence-based weight adaptation to amplify the advantages of the dominant modality. We conduct experiments on three vehicle datasets which with different challenges respectively and achieve state-of-the-art performance. In future work, we will focus on addressing fusion conflicts in complex scenarios with imbalanced modality quality distributions. We will explore other approaches to fusion decoupling, such as introducing causal decoupling, which uses causal graphs to separate modal causal factors from non-causal disturbances, as well as tripartite or multi-subspace decoupling for fusion and decoupling in semantic and other aspects.

VI. REFERENCES SECTION

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